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Atoms and Quantum Physics

The Subatomic World

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Introduction

Everything a person can see, touch, or breathe is built from particles too small to imagine, and the business of those particles is stranger than anything the imagination would have dared propose. An atom is mostly emptiness: a speck of positive charge, a hundred thousand times smaller than the atom as a whole, surrounded by waves of electron stuff that refuse to sit still, refuse to have exact locations, and yet manage to hold the whole of chemistry, color, life, and technology in their arrangement. The science that describes this arrangement is quantum physics, and the partnership between the two subjects is the story of this book: the atom gave quantum physics its first great testing ground, and quantum physics gave the atom, at last, an explanation.

The story is told here as a journey with two halves. The first half follows the atom from its origin as a philosophical guess in ancient Greece, through the chemistry that made it a scientific necessity, to the experiments that broke it apart and found it made of smaller things still: electrons, then a nucleus, then protons and neutrons. The second half follows the quantum revolution that the broken atom forced into being: the discovery that energy comes in grains, that light is both wave and particle, that electrons are waves that can be everywhere at once until looked for, and that the whole grand structure of matter rests on rules that have no counterpart in daily life.

No equations appear in these chapters, because none are needed; the ideas of quantum physics can be carried in plain sentences, and the formal treatments rarely concede what that way of carrying them buys: with the arithmetic set aside, the story becomes a human one: a series of astonished people, each holding a piece of the puzzle, each certain of some things and wrong about others, building in forty crowded years a picture of matter so complete that the everyday world has never needed another. The chapters end where that picture ends: with the lasers, chips, scanners, and clocks that the quantum atom made possible, and with the deep lesson it taught, that the world at bottom is not made of tiny billiard balls but of probabilities, waves, and rules, and that it works anyway, magnificently.

A short guide to what lies ahead, for readers charting a course. The opening three chapters carry the atom from philosophy to pieces: the ancient idea, the electron, and the nucleus. The middle chapters follow the quantum answer to the broken atom, from the grains of light through the Bohr jumps to the wave equation and the exclusion rule that builds the periodic table. The later chapters turn inward to the nucleus and its bargain of energy, and outward to the technologies the finished atom made possible, closing with the atom's standing bequest. The atom itself is the through-line, every chapter stands complete in itself, and a reader chasing a single thread,, the quantum ideas, the nuclear story, or the devices, can follow it without losing the rest.

Chapter 1: The Uncuttable Idea

The atom began as an argument, not an observation. In the Greek cities of the fifth century before the common era, a line of thinkers that included Leucippus and his famous student Democritus proposed that the world must ultimately be made of indivisible units, particles too small to cut, from which everything else was assembled by combination. The reasoning was philosophical: matter appears continuous, but endless divisibility seemed to them absurd, so there must be a floor beneath the knife. They named the floor atomos, uncuttable, and imagined the units as differing only in shape and arrangement, an intuition so far ahead of its evidence that it had to wait twenty-three centuries for confirmation, and which arrived, when it finally did, not by argument but by weighing, mixing, and measuring.

The confirmation, when it came, arrived from an unexpected direction, not from physics but from the laboratory bench of a modest English schoolmaster named John Dalton, who studied the way substances combine. Chemists had established that elements join in fixed proportions, water always the same marriage of two gases, and Dalton, weighing the combinations carefully in the first years of the nineteenth century, found a pattern in the numbers: substances combine in ratios of small whole numbers, as though the units of each substance came in discrete packages. If the packages were real, the ratios followed automatically. His proposal, that each element is made of identical, indivisible atoms distinguished by weight, turned the ancient guess into a working scientific theory, gave chemistry a periodic arithmetic of its own, and made the atom, for the first time, a matter of measurement rather than argument.

The atom's first century as science was a triumph of usefulness and an embarrassment of ignorance at once. The atomic hypothesis worked: it predicted the weights of compounds, organized the elements, and by mid-century had been woven into physics through the kinetic theory of gases, which showed that heat itself is just the motion of invisible particles. But nobody knew what an atom was. Whether it was truly indivisible, whether it had structure, whether the wide variety of matter could really be reduced to some ninety fundamental units: all of it remained open, and a sizable school of respectable scientists, led by the chemist who first systematized the periodic table's precursor patterns, resisted the whole idea as an unprovable convenience, a bookkeeping fiction useful for calculation but not to be taken literally.

The resistance had a point, in the strict terms of its day, and its eventual defeat is the doorway to this whole book. What settled the atom's reality was not more chemistry but new instruments: the Crookes tube and its glowing rays, which began to leak the atom's parts in the 1870s and 1880s, and the radioactive minerals whose faint, ceaseless emanations showed that certain atoms were not stable structures at all but unstable things that broke apart on their own schedule. By 1900, the indivisible unit of matter was visibly coming apart in laboratories across Europe, and the twentieth century opened with the atom's interior, which the ancients had pronounced uncuttable, as the most crowded research frontier in science.

The anti-atomists made one last stand, and its defeat is worth a paragraph because it sealed the subject. Even after chemistry and kinetic theory, a respectable school held that atoms were useful fictions, and their strongest argument was that nobody had ever seen direct evidence of individual particles. The answer came from two directions at once. In 1827 a Scottish botanist had watched pollen grains dance in water, and in 1905 Einstein showed that the dance could be predicted exactly if the water itself were a crowd of jostling molecules, each collision nudging the visible grain. The French physicist Jean Perrin then measured the dancing grains with exquisite patience and found the predictions confirmed in every detail, extracting from pollen the same count of molecules per gram that other methods gave. Within a decade the last holdouts conceded, the constants agreed across a dozen independent techniques, and the atom passed from hypothesis to measured fact, settled in the journals a full generation before anyone saw a picture of one.